

TrES-3b

R-2833

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140807 · created 2026-09-05T23:45:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456876.6534 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.056
Transit depth (Rp/Rs)^2	1.16 +/- 1.2 [%]
Orbital Inclination (inc)	82.37 +/- 2.81
Airmass coefficient 1 (a1)	0.584 +/- 0.096
Airmass coefficient 2 (a2)	0.52 +/- 0.56
Scatter in the residuals of the lightcurve fit is	16.36 %
Best Comparison Star	#3 - [336, 378]
Optimal Aperture	1.54
Optimal Annulus	3.77
Transit Duration (day)	0.033 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.056 vs 0.16309 (deviation 0.98 σ)
Duration fit vs published	0.033 d vs 0.0567 d (deviation 1.39 σ)
Mid-time O-C	-2.9 min
Depth SNR	1.9
Residual scatter	16.36 %

Light curve

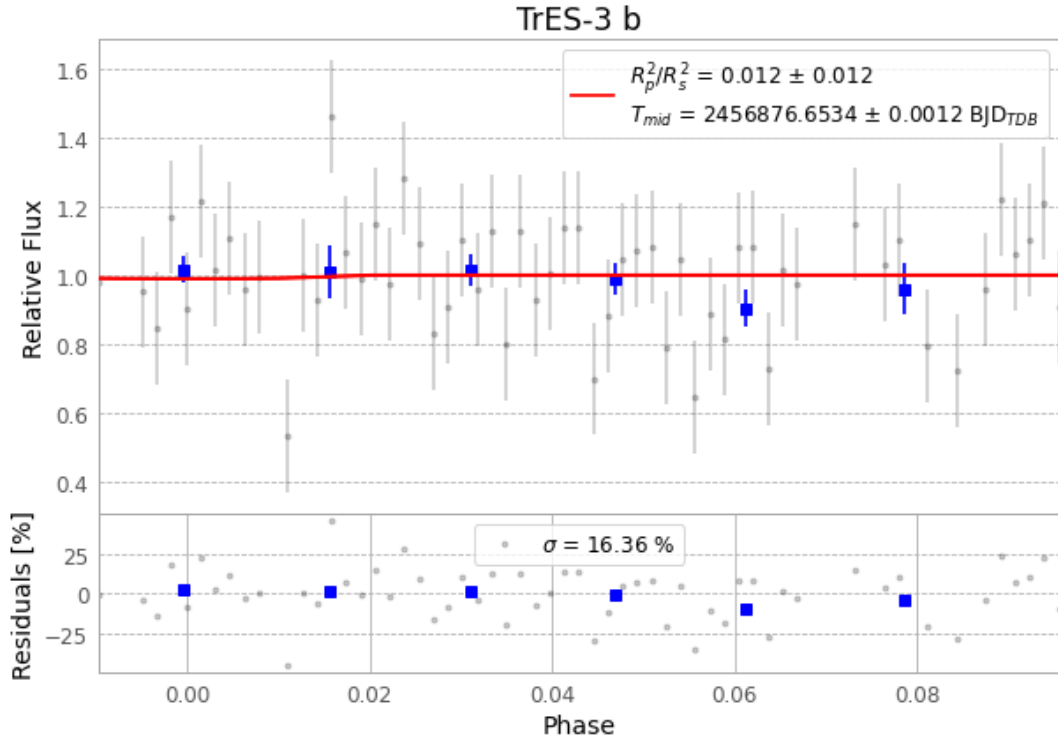


Figure 1 — FinalLightCurve_TrES-3 b_2014-08-06.png (SHA-256 4891d5bd86840a77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d0617c61e21d74cf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3140807031816.FITS → TRES-3140807063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140807.log (acc2a120628e...), FinalLightCurve_TrES-3 b_2014-08-06.png (4891d5bd8684...), FinalLightCurve_TrES-3 b_2014-08-06.csv (5176f7c6cc9f...)

Compute resources

Wall time 202.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:45Z.